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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte(Deemed to be University)
Seventh Semester B.Tech (CBCS) Degree Examinations

December 2025
CS3232-1 – DATA MINING
(For CS)

Duration: 3 Hours

Max. Marks:100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. Which type of data is represented in the form of tables with rows and columns?
A) Object-oriented databases B) Relational databases
C) Transactional databases D) Spatial databases
2. Time-related or sequential data such as stock price movements or sensor data are categorized as
A) Transactional data B) Temporal data
C) Object-oriented data D) Multimedia data
3. Discovering customer segments with similar behavior is an example of
A) Clustering B) Classification
C) Regression D) Association
4. Choosing interesting patterns based on usefulness, novelty, or validity is called
A) Outlier detection B) Pattern evaluation
C) Classification D) Data fusion
5. Attributes that can be ordered but not measured are called
A) Nominal attributes B) Interval-scaled attributes
C) Ratio-scaled attributes D) Ordinal attributes
6. Which of the following distance measures is commonly used for numeric data?
A) Cosine similarity B) Jaccard coefficient
C) Euclidean distance D) Hamming distance
7. The Jaccard coefficient is used to measure similarity between
A) Binary or categorical attributes B) Continuous valued attributes
C) Ratio-scaled attributes D) Ordinal attributes
8. Which of the following is a common data quality problem?
A) Missing values B) Properly formatted data
C) Accurate labels D) Large dataset size
9. A common method for dimensionality reduction is
A) Principal Component Analysis (PCA) B) Data merging
C) Histogram equalization D) Regression
10. According to the Apriori property, if an itemset is frequent, then:
A) All its supersets are frequent B) All its subsets are frequent
C) None of its subsets are frequent D) It must have support 100%
11. The main challenge in Apriori is
A) Handling categorical data B) Generating and testing a large number of candidate itemsets
C) Applying to small databases D) Pruning frequent itemsets



12. The main goal of classification is to
 A) Find hidden relationships
 B) Predict categorical class labels
 C) Summarize data
 D) Optimize transactions
13. Accuracy of a classifier is measured using
 A) Error matrix
 B) Confusion matrix
 C) Correlation matrix
 D) Distance matrix
14. Which of the following is a measure used to select the best attribute in decision trees?
 A) Mean square error
 B) Information Gain
 C) Cosine similarity
 D) Variance
15. Rule-based classifiers represent knowledge in the form of
 A) Clusters
 B) Neural networks
 C) Decision trees
 D) If-Then rules
16. The Sequential Covering Algorithm generates rules by
 A) Building decision tree
 B) Learning one rule at a time and removing covered instances
 C) Clustering the dataset
 D) Generating all possible rules and pruning
17. Cluster analysis is an example of
 A) Unsupervised learning
 B) Supervised learning
 C) Regression analysis
 D) Classification
18. Which of the following is not a clustering method?
 A) Partitioning
 B) Hierarchical
 C) K-Means
 D) Decision Tree
19. Hierarchical clustering produces
 A) Flat clusters
 B) Nested cluster tree (dendrogram)
 C) Rules
 D) Centroids
20. In agglomerative hierarchical clustering, the process starts with
 A) All data in one cluster
 B) Each object as a separate cluster
 C) Random centroids
 D) No clusters

PART - B: DESCRIPTIVE ANSWER QUESTIONS

- | Unit – I | | Marks | BT* | CO* | PO* | | | | | | | | | | | | |
|----------|---|-------|--------------|------|--------------------|------|--------------------|------|--------------|------|-----------------|------|--------------------|--|--|--|--|
| 1. a) | What is data mining? Explain the different kinds of data that can be mined. | 8 | L*2 | 1 | 1 | | | | | | | | | | | | |
| b) | What is an attribute? Explain the different types of attribute with an example. | 8 | L2 | 1 | 1 | | | | | | | | | | | | |
| 2. a) | What is dissimilarity matrix? Explain the methods used to find the dissimilarity of numerical data. | 8 | L2 | 1 | 1 | | | | | | | | | | | | |
| b) | What is the purpose of data cleaning? Explain the different methods used to handle the missing values in attributes. | 8 | L2 | 1 | 1 | | | | | | | | | | | | |
| 3. a) | A database has five transactions. Let min sup = 60% and min conf = 80%. | | | | | | | | | | | | | | | | |
| | <table border="1"> <thead> <tr> <th>TID</th> <th>Items bought</th> </tr> </thead> <tbody> <tr> <td>T100</td> <td>{M, O, N, K, E, Y}</td> </tr> <tr> <td>T200</td> <td>{D, O, N, K, E, Y}</td> </tr> <tr> <td>T300</td> <td>{M, A, K, E}</td> </tr> <tr> <td>T400</td> <td>{M, U, C, K, Y}</td> </tr> <tr> <td>T500</td> <td>{C, O, O, K, I, E}</td> </tr> </tbody> </table> | TID | Items bought | T100 | {M, O, N, K, E, Y} | T200 | {D, O, N, K, E, Y} | T300 | {M, A, K, E} | T400 | {M, U, C, K, Y} | T500 | {C, O, O, K, I, E} | | | | |
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| T400 | {M, U, C, K, Y} | | | | | | | | | | | | | | | | |
| T500 | {C, O, O, K, I, E} | | | | | | | | | | | | | | | | |
| | Apply Apriori algorithm to find all frequent item sets. | 5 | L3 | 2 | 1 | | | | | | | | | | | | |
| b) | Explain the FP-growth algorithm. | 6 | L2 | 2 | 1 | | | | | | | | | | | | |
| c) | Explain how the association rules are generated from the frequent item sets? | 5 | L2 | 2 | 1 | | | | | | | | | | | | |

Unit – II

4.	a)	Explain the data classification process with a neat diagram.	6	L2	3	1
	b)	Explain the following				
		(i) Decision tree				
		(ii) Supervised learning				
		(ii) Unsupervised learning				
		(iv) Attribute Selection Measure				
		(v) Gini Index	5	L2	3	1
	c)	Explain how the information gain can be computed for an attribute that is continuous-valued?	5	L2	3	1
5.	a)	Explain the working of naïve Bayesian classifier.	8	L2	3	1
	b)	Explain the rule based classifier with an example.	8	L2	4	1
6.	a)	Explain the sequential covering algorithm for rule induction.	8	L2	4	1
	b)	Explain the following evaluation measures for the classifier:				
		(i) Accuracy				
		(ii) Error rate				
		(iii) Sensitivity				
		(iv) Specificity	8	L2	4	1

Unit – III

7.	a)	Explain the typical requirements of clustering in data mining.	8	L2	5	1
	b)	Explain the major fundamental clustering methods.	8	L2	5	1
8.	a)	Explain the working of k-means algorithm.	6	L2	5	1
	b)	Explain two types of hierarchical clustering method.	5	L2	5	1
	c)	What is a linkage measure? Explain the types of linkage measures.	5	L2	5	1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
